Analysis of Adult Income Dataset
Statistical Methods for Data Mining

MSC IN APPLIED STATISTICS
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Data Preparation

Data preparation is a critical step before any data mining analysis or machine learning model training. This part of the code focuses on cleaning and preparing the Adult Income dataset, named adult.data. Initially, the data is loaded using the pandas library and read as a dataframe. Additionally, since the file does not contain headers, the columns are manually defined. Missing values are checked for and replaced with the most frequent value of each column to prevent data distortion. Subsequently, to enable the data to be used in machine learning algorithms, all categorical variables are encoded into numerical values using the LabelEncoder class. Furthermore, numerical features are normalized using the MinMaxScaler function. To reduce dimensionality and improve model performance, Principal Component Analysis (PCA) is applied, retaining 95% of the data's variance. The results are as follows:

Principal Component	Explained Variance Ratio	Cumulative Variance
PC1	0.494 (49.40%)	49.40%
PC2	0.216 (21.60%)	71.00%
PC3	0.195 (19.50%)	90.50%
PC4	0.040 (4.00%)	94.50%
PC5	0.026 (2.60%)	97.10%
Total	**0.971**	**97.10%**

Table 1: Explained variance per principal component (Principal Component Analysis)

We see that the features have been reduced from 14 original dimensions to 5 principal components. The first principal component (PC1) explains 49.4% of the total variance, while the first three components cumulatively reach 90.5%. Overall, the 5 components retain 97.1% of the original information, exceeding the 95% target. This means we achieved a 64% dimensionality reduction with only 2.9% information loss, which makes the application of PCA particularly effective for data preparation before training machine learning models.

Feature Selection & Outlier Detection

In this stage, we will perform feature selection and outlier detection. Initially, we identify the most important features that influence the output, namely income, by using the correlation between features through the correlation matrix and analyzing feature importance using a decision tree classifier. Through the correlation matrix, we identify the most significant relationships between features.

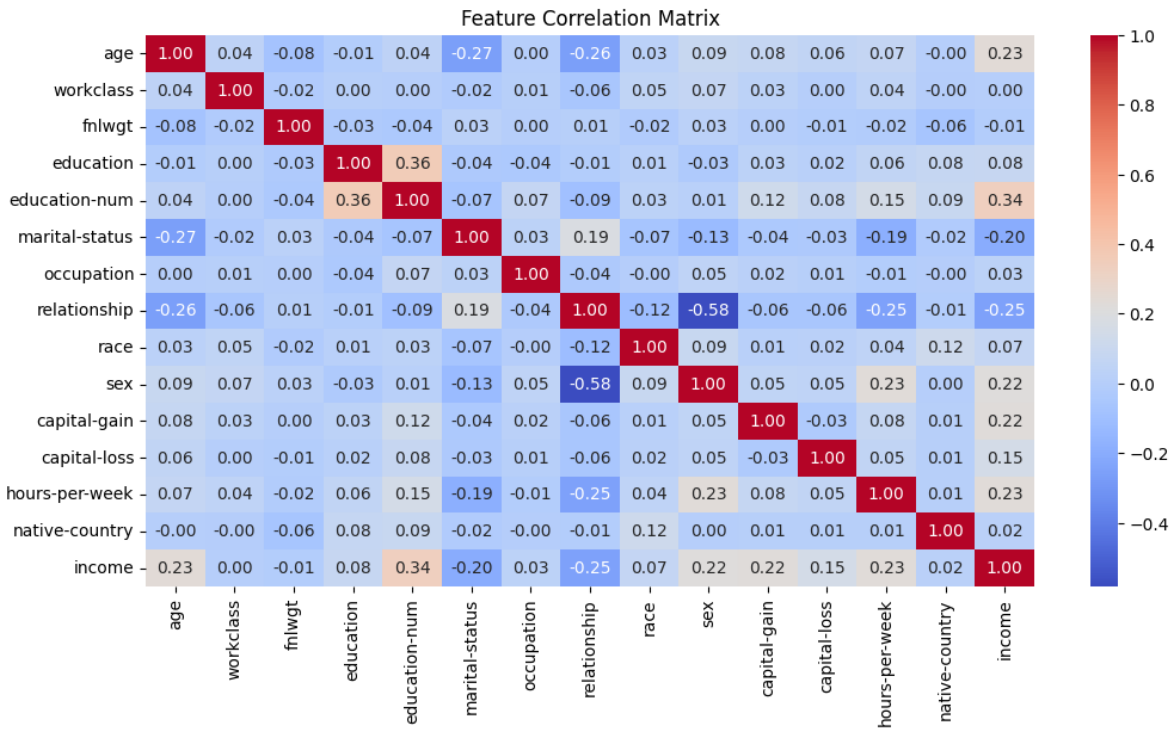

Figure 1: Correlation heatmap of the features (Correlation Heatmap)

Above, we have pairwise correlations between features, and we observe both strong correlations (e.g., sex-relationship = -0.58) as well as weak correlations (e.g., education_num-capital_gain = 0.12). Using the ANOVA statistical test, the five most important features influencing income are: age, education-num, relationship, capital-gain, hours-per-week. Additionally, a Random Forest classifier is employed to evaluate feature importance based on how much each variable affects the model's decisions. This importance is presented in a barplot graph.

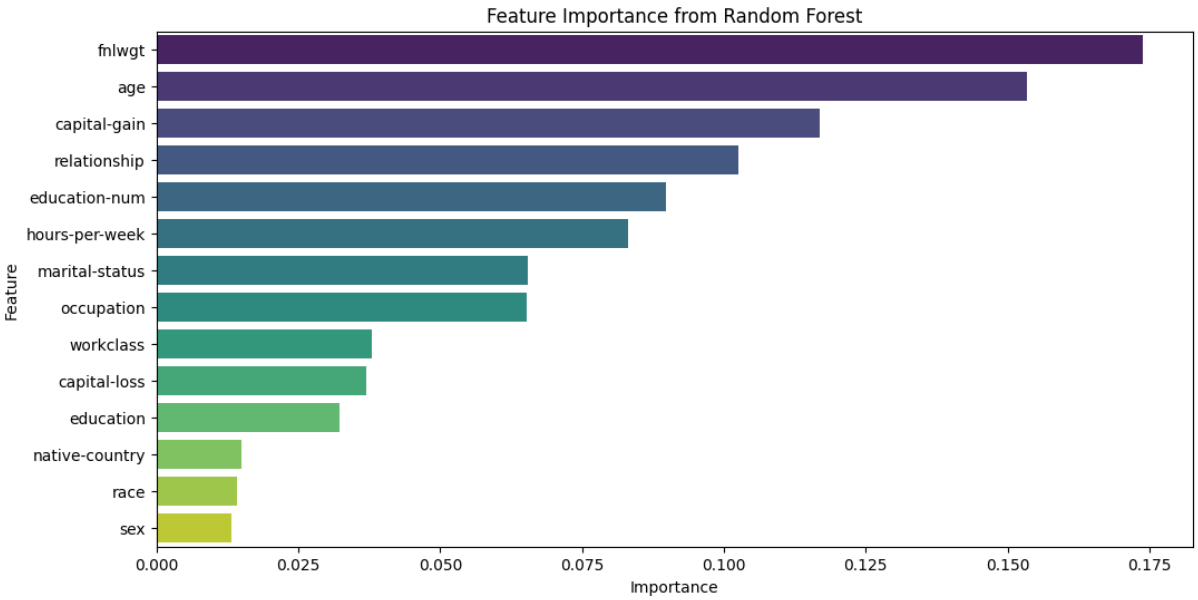

Figure 2: Random Forest - Most important features for income prediction

In addition, we will use the DecisionTreeClassifier function to examine, by removing specific features, how they affect the model's performance and training speed.

Metric	All features (14)	Most important features (5)
Train Accuracy	1.0000	0.9975
Test Accuracy	0.8173	0.7949
Training Time	0.2144 sec	0.1460

Table 2: Comparison of Decision Tree performance with all features vs. 5 most important features

We conclude that with all features we have higher accuracy, and ultimately we will continue using all features. Next, outlier detection is performed using the Z-score method. As you will see in the code, outliers are recorded for each feature separately. We observe that there are numerous outliers, which we will retain because removing them would result in significant information loss from our dataset, and that is not desirable. Below we visualize the outliers.

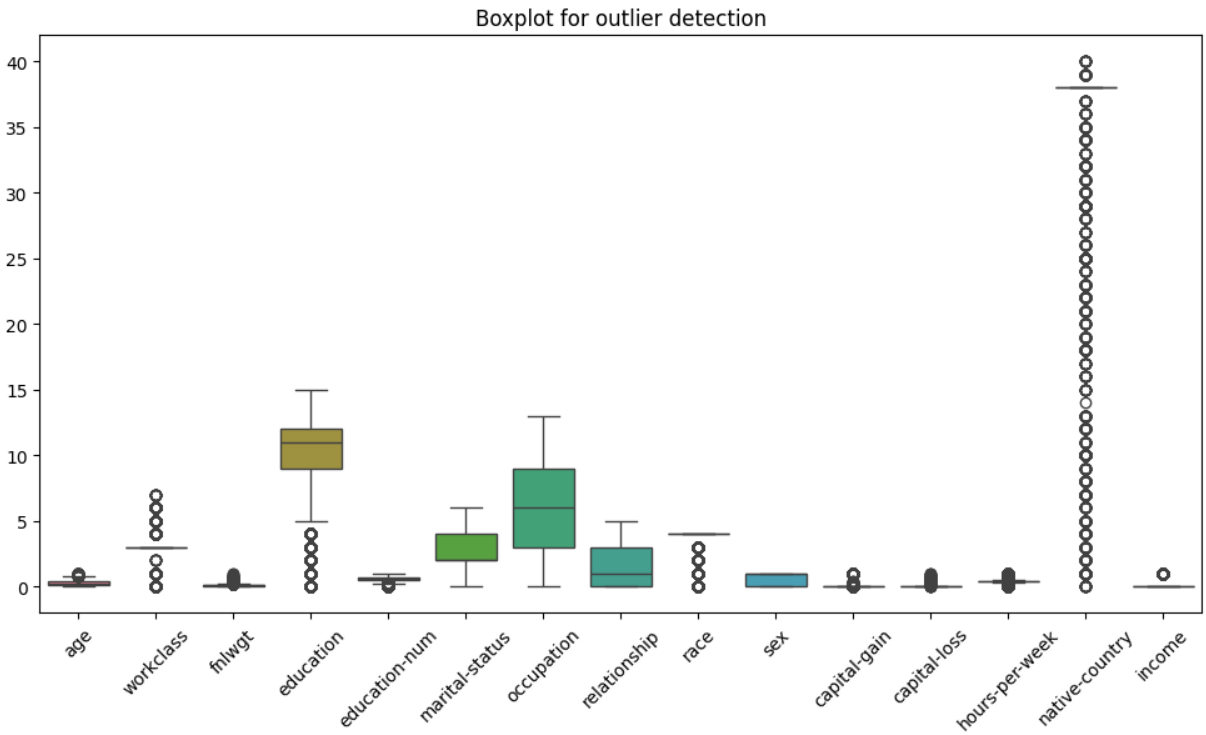

Figure 3: Outlier detection using the Z-score method - Visualization via boxplots for numerical features

The detection of outliers is visualized using boxplots, which illustrate the dispersion of numerical features and potential outliers. The feature selection and outlier detection stage is crucial for the quality of machine learning model results, as it helps reduce complexity, improve prediction accuracy, and avoid overfitting issues. Removing less important features enables the creation of more efficient and interpretable models,

while proper outlier management contributes to prediction stability and better generalization of results.

Clustering with K-Means and Hierarchical Clustering

In this stage, two different clustering methods are applied, K-Means and Hierarchical Clustering, aiming to create two data groups, as we know that the income variable has two categories. Before applying the algorithms, the target variable is removed and Principal Component Analysis (PCA) is applied to reduce the data dimensions and enable their visualization. The first method applied is K-Means, which attempts to find optimal clusters by minimizing the distance of points from the centroid of the cluster to which they belong. The clustering result is stored in the dataset and visualized with a scatter plot of the two principal components. The second method is Hierarchical Clustering, which creates a hierarchical tree where data are gradually merged or split, allowing exploration of clusters at different levels of detail. The result is also stored in the dataset and displayed in a scatter plot. To evaluate the quality of the clusters, the Silhouette Score is calculated, an index that measures how similar data within each cluster are to each other and how differentiated they are from other clusters. The evaluation results for both methods are:

- Silhouette Score for K-Means: 0.645868469721405
- Silhouette Score for Hierarchical Clustering: 0.716790115830591

The higher the Silhouette Score value, the better the clustering. In this specific case, the results appear satisfactory, and the Hierarchical Clustering method groups the data better. Additionally, a dendrogram is created for the Hierarchical Clustering method, which allows understanding of the different relationships between data.

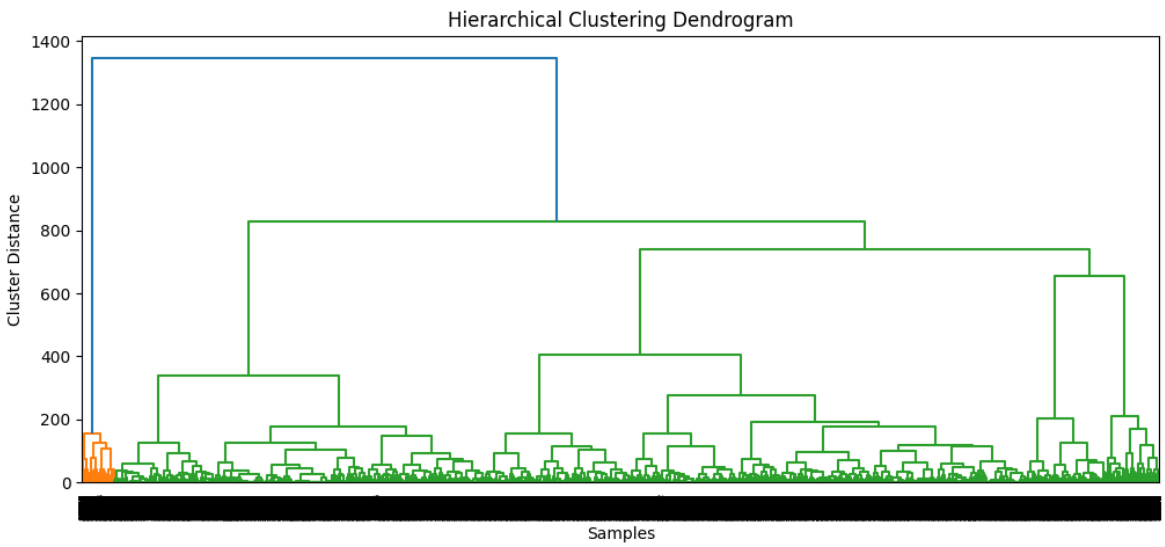

Figure 4: Hierarchical Clustering Dendrogram

Comparing the results with the actual income categories can reveal the extent to which the derived groups correspond to the true classes of the data. This stage provides

insights into the natural grouping of individuals based on their characteristics, which can be utilized in applications such as unsupervised classification or pattern analysis in large datasets.

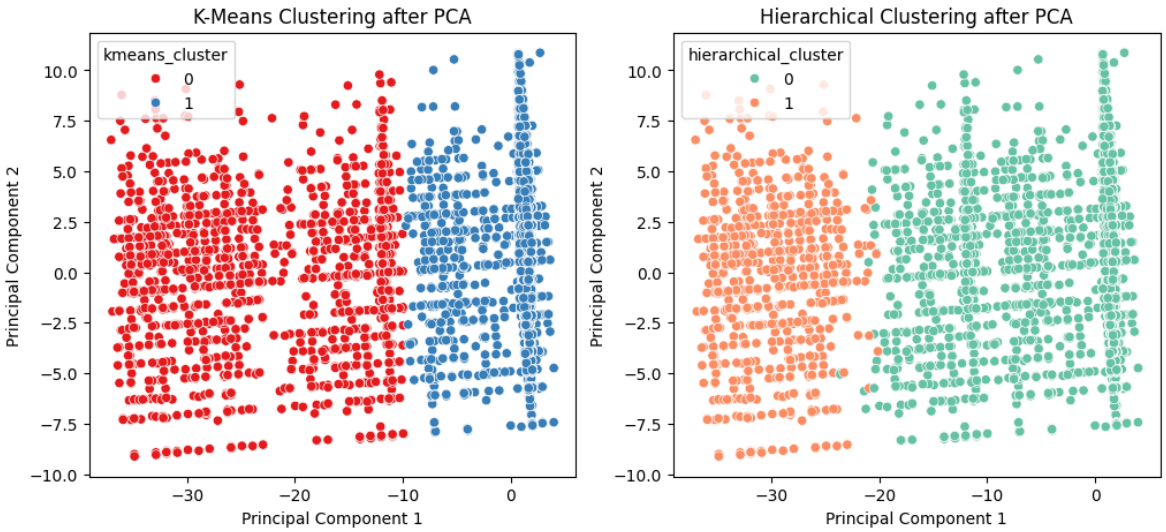

Figure 5: Visualization of K-Means (left) and Hierarchical Clustering (right) clusters

Classification with Scikit-Learn

In this stage, we will train a classification model to predict individuals' income (>50K or ≤50K). We will use at least two different classification algorithms, compare their performance, and evaluate the results by measuring accuracy and other statistical metrics. The steps we will implement include: splitting the data into training and test sets, training two different classification algorithms (Logistic Regression & Random Forest), evaluating the models using metrics such as Accuracy, Precision, Recall, and F1-score, and visualizing the results through a Confusion Matrix.

In this stage, two different classification algorithms are trained, Logistic Regression and Random Forest, to predict individuals' income. Initially, the dataset is split into two sets: a training set containing 80% of the data and a test set containing 20% of the data. The data is normalized using StandardScaler to improve model stability.

The first algorithm applied is Logistic Regression, which is one of the most commonly used methods for binary classification. The model is trained on the training set and then used to predict values on the test set. From this process, we obtain the following results:

Class	Precision	Recall	F1-score	Support
0	0.85	0.94	0.89	4942
1	0.72	0.46	0.56	1571
Accuracy			0.83	6513
Macro Avg	0.78	0.70	0.73	6513
Weighted Avg	0.81	0.83	0.81	6513
Accuracy	0.8259 (82.59%)			

Table 3: Logistic Regression Evaluation Results

The model has an accuracy of 82.59%, meaning it correctly predicts the class in 82.59% of cases. It also performs well on class 0, while its performance on class 1 is lower.

The second algorithm is Random Forest, which consists of multiple decision trees and has the ability to identify more complex patterns in the data, yielding the following results:

Class	Precision	Recall	F1-score	Support
0	0.89	0.93	0.91	4942
1	0.75	0.64	0.69	1571
Accuracy			0.86	6513
Macro Avg	0.82	0.78	0.80	6513
Weighted Avg	0.86	0.86	0.86	6513
Accuracy	0.8609 (86.09%)			

Table 4: Random Forest Evaluation Results

The model has an accuracy of 86.09%, meaning it correctly predicts the class in 86.09% of cases. Additionally, Random Forest outperforms Logistic Regression across all metrics. Performance on class 1 is significantly better, with an F1-score of 0.69 for class 1, which is considerably higher than the 0.56 achieved by Logistic Regression. Therefore, we conclude that Random Forest is the better model for this classification problem. For a better understanding of the models' performance, a confusion matrix is created and visualized, showing the correct and incorrect predictions made by each model. Confusion matrices help identify potential classification issues, such as class imbalance.

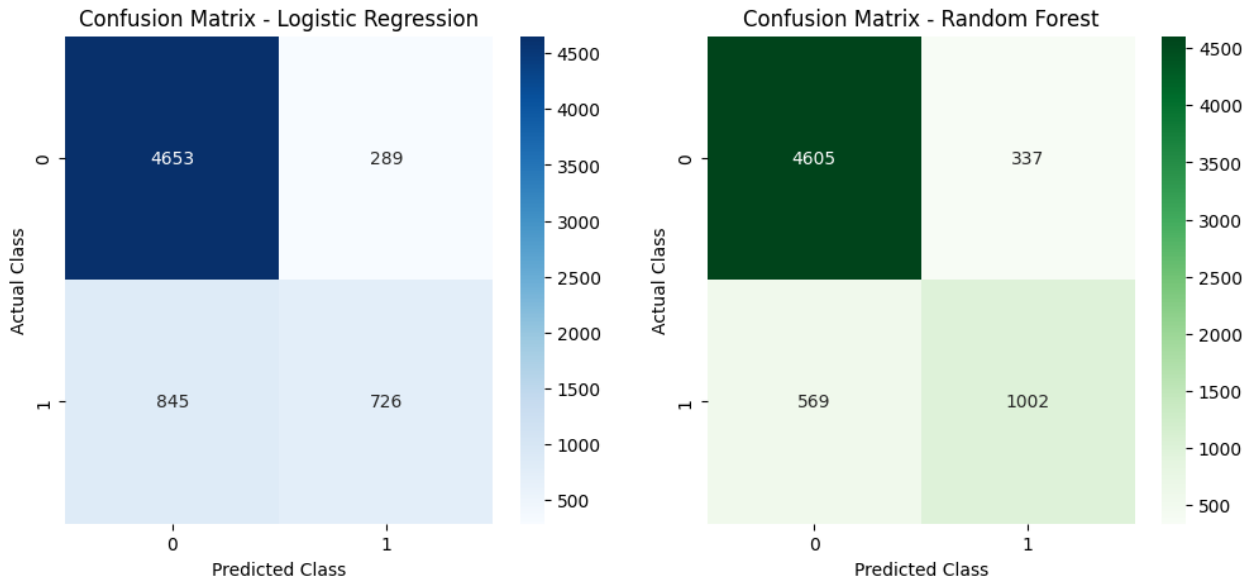

Figure 6: Confusion Matrices of Logistic Regression (left) and Random Forest (right) models

Classification with Neural Networks (Keras/TensorFlow)

In this stage, we will train an MLP (Multilayer Perceptron) neural network using Keras with TensorFlow to predict individuals' income. The dataset is split into training (60%), validation (10%), and test (30%) sets, in order to maintain a separate set for the final evaluation of the model. To stabilize training, the features are normalized using StandardScaler. The neural network consists of three layers: an input layer with 64 neurons, a hidden layer with 32 neurons, and an output layer using the sigmoid activation function, as this is a binary classification task. Additionally, Dropout Layers are introduced to prevent overfitting. Binary crossentropy is selected as the loss function, and Adam with a learning rate of 0.001 is used as the optimizer. The model is trained for 30 epochs with a batch size of 32. At the end of training, evaluation is performed on the test set, where the model's predictions are compared with the actual income categories. The model's performance is assessed by measuring accuracy, recall, precision, and F1-score through the classification report.

Class	Precision	Recall	F1-score	Support
0	0.87	0.94	0.91	7424
1	0.76	0.56	0.65	2345
Accuracy			0.85	9769
Macro Avg	0.82	0.75	0.78	9769
Weighted Avg	0.85	0.85	0.84	9769
Accuracy	0.8523 (85.23%)			

Table 5: Neural Network Evaluation Results

In parallel, the evolution of the loss function and accuracy over the epochs is visualized, in order to determine whether the model is overfitting or continues to improve with additional training epochs.

Figure 7: Visualization of trainning progress - Loss (left) and Accuracy (right) for train and validation sets

Finally, the confusion matrix is created, which illustrates the correct and incorrect classifications of the model on the test set. This stage is particularly important as it allows for comparison of the results with previous models, such as Logistic Regression and Random Forest, and provides insights into whether neural networks offer an improvement in classifying individuals' income.

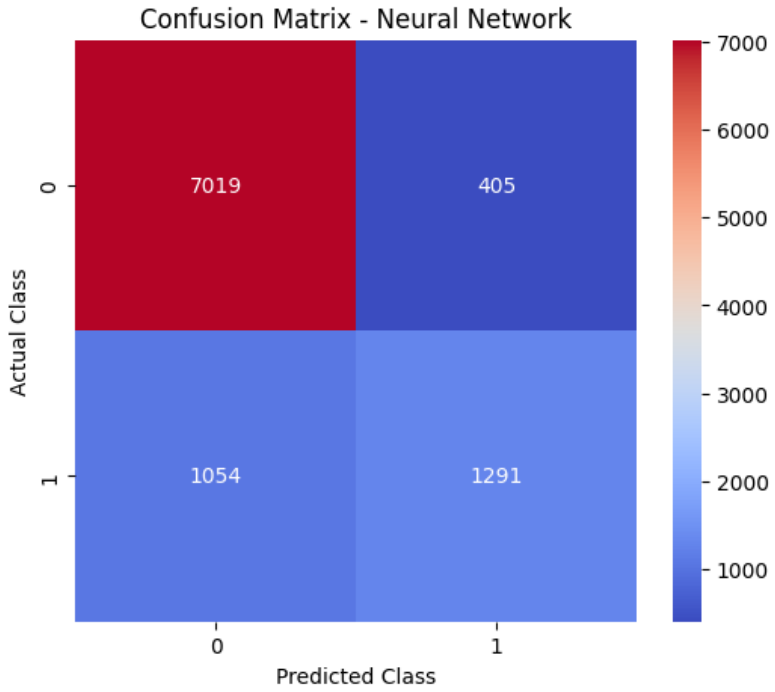

Figure 8: Confusion Matrix of the Neural Network